

Question 26

A version control system (VCS) is a tool that tracks and manages changes to files and source code over time, allowing teams to collaborate and recover earlier versions if needed.

Git is a free open source distributed version control system that allows every user to have a complete copy of the project's history on their own machine.

It is widely used by analysts and developers because:

- ✦ It allows for collaboration where multiple users can work on the same file at the same time.
- ✦ It is a safety net that allows users to go back to previous state if they make a mistake.
- ✦ It allows for traceability hence maintaining a detailed history of who made the changes.

Core Git Concepts

- Commit - is a save point of a project at a specific moment in time.
- Each commit details the changes that were made.
- Branch - it is a separate version of the codebase that branches off from the main project.
- It allows one to experiment with new features or different models without affecting the main version.
- Once the work is ready, the branch can be merged back into the main branch.